

① (a). point estimation:

- Gives a single value that serves as an estimate of a population parameter.
- Does not reflect the uncertainty or variability in the estimate.
- E.g:- Sample mean

Interval Estimation:

- Gives a range of values used to estimate a population parameter
- Reflects uncertainty by providing a range, with a confidence interval
- E.g:- 95% C.I for mean

$$(b). (i). \bar{X} = \frac{\sum X}{n} = \frac{1111}{20} = 55.55$$

$$\sigma = \sqrt{\frac{1}{(n-1)} (\sum X_i^2 - n\bar{X}^2)} = \sqrt{\frac{1}{19} (69909 - 20 \times 55.55^2)}$$
$$= 20.776$$

$$\begin{aligned} 95\% \text{ C.I} &= \bar{X} \pm t_{\alpha/2, n-1} \left(\frac{\sigma}{\sqrt{n}} \right) \\ &= 55.55 \pm t_{0.025, 19} \left(\frac{20.766}{\sqrt{20}} \right) \\ &= 55.55 \pm 2.093 \left(\frac{20.766}{\sqrt{20}} \right) \\ &= 55.55 \pm 9.719 \\ &= \underline{\underline{(45.831, 65.269)}} \end{aligned}$$

We are 95% confident that the population mean time required to complete the task lies between 45.831 and 65.269 minutes.

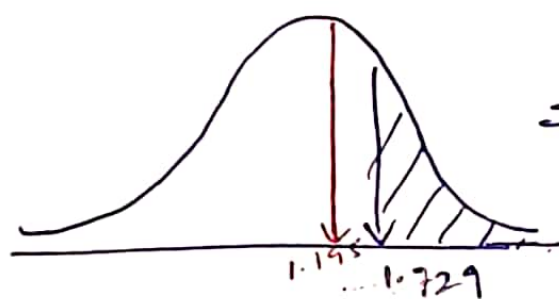
(ii). Hypothesis :

$$H_0 : \mu = 50 \quad \text{vs} \quad H_1 : \mu > 50$$

$$\begin{aligned} \text{Test statistics} &= \frac{\bar{X} - \mu}{s/\sqrt{n}} \\ &= \frac{55.55 - 50}{20.776/\sqrt{20}} \\ &= 1.195 \end{aligned}$$

$$\text{Critical value} = t_{19, 0.05} = 1.729$$

Decision Rule :



$\Rightarrow$ Do not reject H_0

Conclusion :

There is not enough evidence to conclude that the expected time take to complete the task is greater than 50 minutes at 5% sig. level.

②. a). (i). Type I error - Rejecting a null hypothesis when it is true

Type II error - Failing to reject a null hypothesis when it is false.

(ii). Independent samples - The samples are unrelated and drawn from two different populations or groups of subjects.

Dependent samples - The samples are related or paired in some way.

b). Hypothesis :

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

or

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

or

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \neq 0$$

Student Name	Commerce Marks (A)	Mathematics Marks (B)	$d = A - B$	$(d_i - \bar{d})^2$
A	51	67	-16	174.24
B	42	54	-12	84.64
C	56	53	3	33.64
D	58	49	9	139.24
E	38	47	-9	38.44
F	44	35	9	139.24
G	62	30	32	1211.04
H	39	77	-38	1239.04
I	48	57	-9	38.44
J	57	54	3	33.64
			$\sum d = -28$ $\bar{d} = -2.8$	$\sum (d_i - \bar{d})^2 = 3131.6$

$$\bar{d} = \frac{-28}{10} = -2.8$$

$$S_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{(n-1)}} = \sqrt{\frac{3131.6}{9}} = 18.65$$

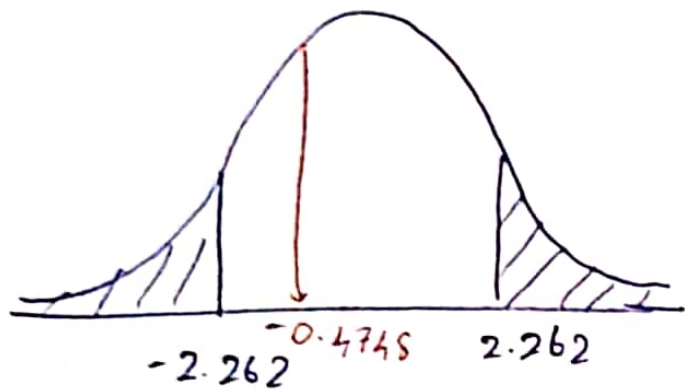
Test Statistics :

$$t = \frac{\bar{d}}{S_d / \sqrt{n}} = \frac{-2.8}{18.65 / \sqrt{10}} = -0.4748$$

Critical value :

$$t_{0.025, 9} = 2.262$$

Decision rule:



$\Rightarrow$ Do not reject H_0 .

Conclusion:

We have enough evidence to conclude that the expected Commerce mark and Mathematics mark in grade 10 are equal for a randomly selected student at 5% significance level.